

Computer Organization & Assembly Language

Lab 10

Lab Instructor: Mr. Tariq Mehmood Butt

Topic: XLAT Instruction

LAB TASKS (60 marks)

Task 1:

Write an assembly program that takes an alphabet character as input. Use XLAT instruction with a translation table to convert it to opposite case (lowercase to uppercase and vice-versa) and display the result.

Task 2:

Implement a simple ROT13 cipher for a single character. Use XLAT instruction with a pre-defined ROT13 translation table for encryption. Display the encrypted result.

ROT13 replaces a letter with the one 13 places down the alphabet. For example, the letter **A** is replaced with **N**, **B** with **O**,, **a** with **n**, and **b** with **o**, and so on.

Task 3:

Write an assembly program that takes an alphabet character as input. Use XLAT instruction to check if the input character is vowel. Display 'Y' for vowel and 'N' for consonant.

HOME TASK (40 marks)

Write a procedure **take_input** which is capable of getting input in any format (binary, octal, decimal or hexadecimal). The procedure should accept the input format (i.e. 2, 8, 10, 16) in **BX** and return the input in **AX**.

Write a procedure **display_output** which is capable of displaying the content of a 16-bit register (use **AX** to pass value) in any format (binary, octal, decimal or hexadecimal). The procedure should accept the output format (i.e. 2, 8, 10, 16) in **BX**.

Note: Both procedures should work for all the formats using the same general logic (not separate logic for each format).

Hint:

- Use **MUL** instruction to build the number in **take_input** procedure.
- Use **DIV** instruction to get **remainder** in **display_output** procedure.

Show complete working in the **main** procedure.

Implement main procedure to:

- 1) Take input in **binary** and display output in **octal, decimal and hexadecimal**.
- 2) Take input in **octal** and display output in **binary, decimal and hexadecimal**.
- 3) Take input in **decimal** and display output in **binary, octal and hexadecimal**.
- 4) Take input in **hexadecimal** and display output in **binary, octal and decimal**.

